

Mathis Azéma

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Research profile

I am a PhD candidate in optimization under uncertainty, supervised by Vincent Leclère and Wim van Ackooij. A central question in my doctoral research is how distributionally robust optimization can be applied to unit commitment and what it can offer compared with stochastic and robust approaches, both computationally and in terms of out-of-sample performance. More broadly, I study decomposition methods for stochastic, robust and distributionally robust optimization, particularly for mixed-integer problems arising in power systems. I work on strengthening Benders and SDDP cuts, developing an infinite-horizon SDDP method, and studying Lagrangian reformulations for robust optimization with mixed-integer recourse. These projects include collaborations with Wolfram Wiesemann and Bernardo da Costa.

Education

PhD in applied mathematics – CERMICS, École Nationale des Ponts et Chaussées

April 2024 – September 2027, Champs-sur-Marne, France.

Supervisors: Vincent Leclère and Wim van Ackooij.

Master's degrees – 2024

Operations Research (MPRO), Conservatoire National des Arts et Métiers, Paris, France.

Optimization and Data Science, ENSTA Paris, Palaiseau, France.

Engineering degree – École Polytechnique, 2020–2024

Palaiseau, France.

Ranked 10th out of 425 students.

First-year master's research internship – Polytechnique Montréal (GERAD)

April–August 2023, Montréal, Canada.

Research on electric bus assignment with parking constraints, using mixed-integer linear programming and constraint programming.

Journal articles

1. Distributionally robust unit commitment models: Theory and numerical results.

M. Azéma, V. Leclère, W. van Ackooij.

Forthcoming in *Electric Power Systems Research*, vol. 263, 113754 (2027). doi:10.1016/j.epsr.2026.113754

We developed a unified Benders framework to compare stochastic, robust and distributionally robust unit commitment models, using difference-of-convex methods to address nonconvex Wasserstein separation problems. We evaluated computational efficiency and out-of-sample performance, finding that the L^2 norm achieved the best empirical performance among the norms tested.

2. Stronger cuts for Benders' decomposition for stochastic Unit Commitment Problems based on interval variables.

M. Azéma, V. Leclère, W. van Ackooij.

Mathematical Programming Computation, 2026. doi:10.1007/s12532-026-00334-1

We proved that an extended formulation using operating-interval variables can yield stronger Benders cuts than the standard formulation for two-stage stochastic unit commitment. Fast cut coefficients based on weak duality improve computational performance on large-scale stochastic unit commitment benchmarks.

3. A mixed-integer linear programming model for the electric bus assignment problem with parking constraints.

M. Azéma, G. Desaulniers, J. E. Mendoza, G. Pesant.

Accepted for publication in *Public Transport*.

We developed a mixed-integer linear programming formulation for electric bus assignment with parking constraints, providing an alternative modeling approach to constraint programming.

Manuscripts in preparation

1. **Vector and Scalar Lagrangian Reformulations for Mixed-Integer Recourse under Binary Uncertainty.**

M. Azéma, V. Leclère, W. van Ackooij, W. Wiesemann.
Manuscript in preparation.

We studied two-stage robust optimization with mixed-integer recourse, where solving the max–min worst-case problem is a major computational bottleneck. Under binary uncertainty, we compared exact Lagrangian reformulations that transform this problem into a single-level mixed-integer linear program. We established relationships between the strength of their continuous relaxations and evaluated their computational performance on robust staff rostering instances.

2. **Stronger cuts for stochastic dual dynamic programming for multistage stochastic unit commitment problems based on interval variables.**

M. Azéma.
Manuscript in preparation.

We proved that operating-interval formulations can yield stronger SDDP cuts for multistage stochastic unit commitment, where generation decisions adapt to demand as it is observed.

3. **Relative Value Iteration for Infinite-Horizon SDDP: Application to Hydroelectric Problem (RVS-DDP).**

M. Azéma, B. F. P. da Costa, F. Durand, V. Leclère.
Manuscript in preparation.

We studied infinite-horizon stochastic hydroelectric planning, where standard SDDP methods face numerical difficulties when the discount factor is close to one. We proposed a modified SDDP algorithm inspired by relative value iteration, using additive shifts to improve the relevance of generated cuts and accelerate convergence. We proved the theoretical convergence of the proposed algorithm and evaluated its computational performance through numerical experiments on long-term hydroelectric planning.

Conference publications

1. **A Constraint Programming Model for the Electric Bus Assignment Problem with Parking Constraints.**

M. Azéma, G. Desaulniers, J. E. Mendoza, G. Pesant.
CPAIOR 2024, LNCS 14742, pp. 17–33. doi:10.1007/978-3-031-60597-0_2

We developed a constraint programming model for electric bus assignment that accounts for parking constraints.

Selected talks

- **PSCC 2026**, Limassol, Cyprus — Distributionally Robust Unit Commitment Models: Theory and Numerical Results.
- **SIAM Conference on Optimization 2026**, Edinburgh, United Kingdom — Relative Value Iteration for Infinite-Horizon SDDP: Application to Hydroelectric Planning.
- **ICSP 2025**, Champs-sur-Marne, France — Solving a Two-Stage Distributionally Robust Unit Commitment Model Using Wasserstein Distance.
- **EUROPT 2025**, Southampton, United Kingdom — Solving a Convex Quadratic Maximization Problem Arising in Distributionally Robust Optimization.
- **EURO 2025**, Leeds, United Kingdom — Enhanced Benders' Decomposition for Stochastic Unit Commitment Using Interval Variables.
- **ROADEF 2025**, Paris, France — Solving a Convex Quadratic Maximization Problem Arising in Distributionally Robust Optimization.

Award

ROADEF Master's Thesis Award (2023 edition) – Second prize, awarded in 2024, for *Electric Vehicle Assignment Problem with Parking Constraints*.

Teaching experience

Teaching assistant – École Nationale des Ponts et Chaussées, 2024–2026 (approximately 45 hours in total)

- Led exercise sessions for *Introduction to Optimization*, first-year engineering students.
- Led exercise sessions for *Convex Optimization*, second-year engineering students.